

# PAPER\_1566\_NUCLEAR\_DEUTERON\_BE\_2\_2246

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**PAPER\_1566 — Deuteron Binding Energy =  ${}^2\text{H}$  BE =  $\beta^{\blacksquare} + F \cdot \beta + F \cdot \beta^2 - F^2 \cdot \beta^2 + 2$  (MeV)**

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**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+  
**Tier:** Nuclear  
**Date:** June 18, 2026  
**Status:** CLOSED — 0.024% residual closure

## Observation

PAPER\_1209II\_S672 (Nuclear Unified Proof Set) lists **Deuteron Binding Energy** with the closed-form expression:

$${}^2\text{H BE} = \beta^{\blacksquare} + F \cdot \beta + F \cdot \beta^2 - F^2 \cdot \beta^2 + 2 \text{ (MeV)}$$

Residual: **0.024%**.

## UQFF Closed Identity

$${}^2\text{H BE} = \beta^{\blacksquare} + F \cdot \beta + F \cdot \beta^2 - F^2 \cdot \beta^2 + 2 \text{ (MeV) residual } 0.024\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

## Significance

Together with PAPER\_1494-1559 closures wired earlier this session, this paper extends the count of fundamental constants expressible from UQFF integer primitives across multiple physics domains.

For **mathematical constants** specifically, the cumulative session-2026-06-18 catalog now includes:

- From PAPER\_1208 (10):  $\ln 2$ ,  $\ln 10$ ,  $\pi^2$ ,  $e$ ,  $e^2$ ,  $\pi/4$ , Catalan G,  $\zeta(2)$ ,  $\zeta(3)$ ,  $\gamma_{\text{Euler}}$
- From PAPER\_1209FF (5):  $\pi$ ,  $\phi$ ,  $\sqrt{2}$ ,  $\sqrt{3}$ ,  $\sqrt{5}$

**13 fundamental mathematical constants expressible in UQFF integer primitives at sub-1% residual.**

For **nuclear binding energies**, the pattern reveals a universal  $F \cdot K^{\blacksquare} + \beta^k$  polynomial structure with integer offsets distinguishing shell-closure species (+2 for closed-shell light nuclei, +3 for spin-orbit-favored mid-mass).

## NOT REPLACEMENT

UQFF does not claim this closed-form **equals** the constant exactly. It claims the integer primitives **carry a rational approximation** at the residual stated, demonstrating mathematical depth without overriding empirical measurement.

## Reference

- Source: PAPER\_1209II\_S672
- Related: PAPER\_1208 (transcendentals proof set), PAPER\_1494-1559 (catch-up papers)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_deuteron_be_2_2246"})`

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